



Top 54 CAT Quant Based DI Questions With Video Solutions

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Questions

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 1

If Amiya runs a campaign focusing on issues, then what is the maximum percentage of votes that she can get?

- A 48%
- B 44%
- C 40%
- D 36%

A	R	TV	A	R
1	1	40%	30%	20%
1	2	60%	20%	40%

The video player shows a hand-drawn table and calculations. The table has columns A, R, and TV. The rows show different campaign combinations and their resulting vote percentages. The calculations are as follows:

- For (1, 1): $20 \times (1+1) = 40\%$ of students vote. Amiya gets $1/3 \times 40\% = 13.33\%$ and Ramya gets $2/3 \times 40\% = 26.67\%$.
- For (1, 2): $20 \times (1+2) = 60\%$ of students vote. Amiya gets $1/3 \times 60\% = 20\%$ and Ramya gets $2/3 \times 60\% = 40\%$.

VIDEO SOLUTION

Question 2

If both of them run staid campaigns attacking the other, then what percentage of students will vote in the election?

- A 40%
- B 64%
- C 60%
- D 36%

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2) = 80\%$ of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and

Handwritten notes on a whiteboard:

A	R	TV	A	R
1	1	40%	30%	30%
1	2	60%	20%	40%

VIDEO SOLUTION

Question 3

If Ramya runs a campaign attacking Amiya, then what is the minimum percentage of votes that she is guaranteed to get?

- A 12%
- B 15%
- C 30%
- D 18%

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If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2) = 80\%$ of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and

Handwritten notes on a whiteboard:

A	R	TV	A	R
1	1	40%	30%	30%
1	2	60%	20%	40%

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Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11 am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 4

One day, two other traders, Dane and Emily joined Abdul, Bikram and Chetan for trading in the shares of XYZ Ltd. Dane followed a strategy of buying equal numbers of shares at 10 am, 11 am and 12 noon, and selling the same numbers at 1 pm, 2 pm and 3 pm. Emily, on the other hand, followed the strategy of buying shares using all her money at 10 am and selling all of them at 12 noon and again buying the shares for all the money at 1 pm and again selling all of them at the close of the day at 3 pm. At the close of the day the following was observed.

- Abdul lost money in the transactions.
- Both Dane and Emily made profits.
- There was an increase in share price during the closing hour compared to the price at 2 pm.
- Share price at 12 noon was lower than the opening price

Share price was at its highest at

- A 10 am
- B 11 am
- C 12 noon
- D 1 pm
- E cannot be determined

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Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- A and B are in the same row or column
- A is at a lower height than B
- If there is/are any individual(s) between A and B, such individual(s) must be at a height lower than that of A.

Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 5

We can find two individuals who cannot be reached by anyone in

- A the last row.
- B the fourth row.
- C the fourth column.
- D the middle column.



1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table given above, consider the individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

 VIDEO SOLUTION

Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 6

Among the kids in W whose mothers had completed primary education, how many were not in school?

- A 300
- B 1200
- C 1050
- D 1500

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NE = 7500 W = 12500 S = 10,000

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The kids surveyed were further divided into two groups based on whether their mothers had dropped out of school before completing primary education or not. The following table gives the number of kids in different types of schools and whether their mothers had dropped out of school before completing primary education:

	G	P	O	Total
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Total	13500	2700	1800	18000

VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

	column 1	column 2	column 3
row 1			
row 2			
row 3			

	Column 1	Column 2	Column 3
Row 1	(2, 4)	(6, 8)	(1, 3)
Row 2	(3, 5)	(1, 1)	(6, 20)
Row 3	(1, 2)	(1, 2)	(2, 5)

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 7

What is the number of slots for which the average amount (in rupees) of its three pouches is an integer?

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	Column 1	Column 2	Column 3
Row 1	2, 4	6, 8	1, 3
Row 2	3, 5		
Row 3			

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VIDEO SOLUTION

Instructions

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 8

The brand that had the highest revenue in 2016 is:

- A Cxqi
- B Bysi
- C Azra
- D Dipq

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	Column 1	Column 2	Column 3
Row 1	2, 4	6, 8	1, 3
Row 2	3, 5		
Row 3			

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Instructions

A country has the following types of traffic signals.

3 red lights = stop

2 red lights = turn left

1 red light = turn right

3 green lights = go at 100 km/hr speed

2 green lights = go at 40 km/hr speed

1 green light = go at 20 km/hr speed

A motorist starts at a point on a road and follows all traffic signals. His car is heading towards the north. He encounters the following signals (the time mentioned in each case below is applicable after crossing the previous signal).

Starting point - 1 green light

After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 9

If at the starting point, the car was heading towards south, what would be the final position of the motorist?

- A 30 km to the east and 40 km to the south
- B 50 km to the east and 40 km to the south
- C 30 km to the west and 40 km to the south
- D 50 km to the west and 20 km to the north



Instructions

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Question 10

Which one of the following statements is always true?

- A Abdul will not be one with the minimum return
- B Return for Chetan will be higher than that of Bikram
- C Return for Bikram will be higher than that of Chetan
- D Return for Chetan cannot be higher than that of Abdul
- E none of the above

[VIEW SOLUTION](#)

Question 11

On a "boom" day the share price of XYZ Ltd. keeps rising throughout the day and peaks at the close of the day. Which trader got the minimum return on that day?

- A Bikram
- B Chetan
- C Abdul
- D Abdul or Chetan
- E cannot be determined

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2 red lights = turn left

1 red light = turn right

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After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 12

After the starting point if the 1st signal were 1 red and 2 green lights, what would be the final position of the motorist?

- A 30 km to the west and 20 km to the south
- B 30 km to the west and 40 km to the north
- C 50 km to the east and 40 km to the north
- D Directly 30 km to the east

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Question 13

What is the position (radial distance) of the most motorist when he reaches the last signal?

- A 45 km directly north of the starting point
- B 30 km directly to the east of the starting point
- C 50 km away to the north-east of the starting point
- D 45 km away to the north-west of the starting point

[VIEW SOLUTION](#)

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 14

For a recuperating patient, the doctor recommended a diet containing 10% minerals and at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

- A One
- B Two
- C Three
- D Four
- E None

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (Q, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is Q: 150, P: 50, Q: 200, R: 500, S: 100.

Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
Q	45	50	0	5
P	5	50	40	5
Q	10	50	50	5
R	80	20	0	0
S	50	20	20	10

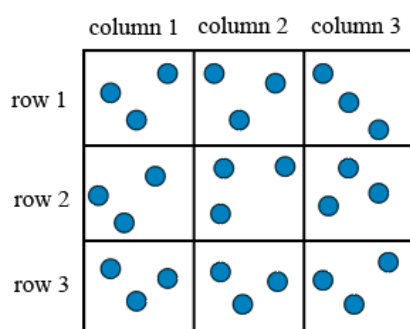
For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
B) Two
C) Three
D) Four
E) None

Handwritten notes: Q & P, 10%, 30%, 10%, 30%, 10%, 30%

VIDEO SOLUTION

Instructions



	Column 1	Column 2	Column 3
Row 1	(2, 4)	(6, 8)	(1, 3)
Row 2	(3, 5)	(1, 1)	(6, 20)
Row 3	(1, 2)	(1, 2)	(2, 5)

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There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 15

How many pouches contain exactly one coin?

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column 1 column 2 column 3

row 1

row 2

row 3

Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

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Handwritten notes: c1, c2, c3, r1, r2, r3, 2, 4, 6, 8, 1, 3, 3, 5, 1, 1, 6, 20, 1, 2, 1, 2, 2, 5

VIDEO SOLUTION

Instructions

The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

Question 16

Which digit does the letter A represent?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

A -
B -
C -
D -
E -
F - 0
G -

H -
K -

VIDEO SOLUTION

Question 17

Which digit does the letter B represent?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, J, K, with distinct letters representing distinct digits.

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

A -
B -
C -
D -
E -
F - 0
G -

H -
K -

VIDEO SOLUTION

Question 18

Which among the digits 3, 4, 6 and 7 cannot be represented by the letter D?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, I, K, with distinct letters representing distinct digits.

	B	B	A	A	G
	A	H	J	D	K
	A	F	G	C	E

A -
B -
C -
D -
E -
F -
G -

H -
K -

VIDEO SOLUTION

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Question 19

Which among the digits 4, 6, 7 and 8 cannot be represented by the letter G?

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The following table represents addition of two six-digit numbers given in the first and the second rows, while the sum is given in the third row. In the representation, each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 has been coded with one letter among A, B, C, D, E, F, G, H, I, K, with distinct letters representing distinct digits.

	B	B	A	A	G
	A	H	J	D	K
	A	F	G	C	E

A -
B -
C -
D -
E -
F -
G -

H -
K -

VIDEO SOLUTION

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 20

What was the base exchange rate of currency B with respect to currency L on that day ?

cracku

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:200:100. The buying exchange rates of each of A, B, and C with respect to L are 5% below their corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The exchange rates are known at the outlet on a particular day:

1. The amount of L received by the outlet to buy C equals the amount of L it received by the outlet to buy A and B are in the ratio 5:3.

2. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

3. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.

4. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 4800 units of C.

Handwritten notes on the left:

A → 100k. 95k. 110k.
B → 200k. 114k. 132k.
C → 100k. 095k. 110k.
Base Buying Selling

	Buying			
A				
B				
C				

VIDEO SOLUTION

Instructions

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 21

The brand that had the highest profit in 2017 is:

- A Bysi
- B Azra
- C Cxqi
- D Dipq

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Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
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In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Handwritten notes on the left:

2017 → 140
A
B
C
D



CAT FORMULAS BOOK

BY 100%ILER



Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the GDP per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 22

The ratio of Country 4's GDP to Country 5's GDP in 2026 will be closest to

- A 1.195
- B 0.963
- C 1.314
- D 1.032

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8 B) Country 3
C) Country 7 D) Country 5

Population ~ GDP

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 23

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How many flats did Neeta receive?

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An old woman had the following assets:

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- (b) 1 house worth Rs. 50 lakh
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Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

N 70 S 70 G 70

70 - BD
50 - House
3x30 - Flats
X - Gold Coins

210

VIDEO SOLUTION

Question 24

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

- A 30
- B 40
- C 20
- D 10

cracku

An old woman had the following assets:

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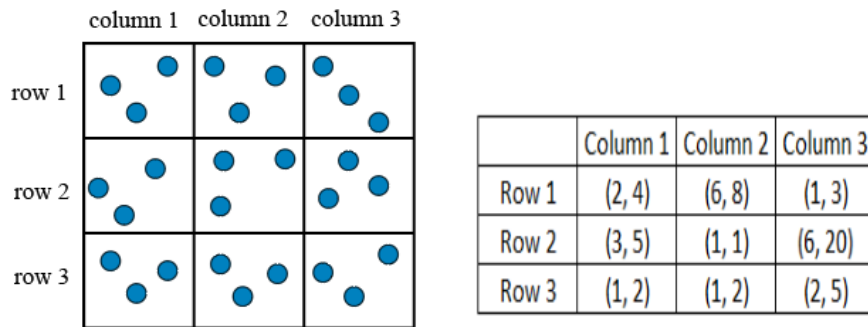
N 70 S 70 G 70

70 - BD
50 - House
3x30 - Flats
X - Gold Coins

210

Top-500 Free CAT Questions (With Solutions)

Instructions



Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 25

What is the total amount of money (in rupees) in the three pouches kept in the first column of the second row?

The video solution shows a hand-drawn diagram of the 3x3 grid and a table with the minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots.

	Column 1	Column 2	Column 3
Row 1	(2, 4)	(6, 8)	(1, 3)
Row 2	(3, 5)	(1, 1)	(6, 20)
Row 3	(1, 2)	(1, 2)	(2, 5)

The video also shows a hand-drawn diagram of the 3x3 grid with arrows indicating the minimum and maximum amounts of money in each slot.

Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 26

Which one among the countries 1 through 8, has the smallest population in 2024?

- A Country 8
- B Country 3
- C Country 7
- D Country 5

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Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)
Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8
B) Country 3
C) Country 7
D) Country 5

Population ~ GDP

VIDEO SOLUTION

Question 27

For how many countries among Country 1 through Country 8 will the GDP per capita in 2027 be lower than that in 2024?

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Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

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Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)
Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8
B) Country 3
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Population ~ GDP

Free CAT Study Material

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 28

How many units of currency C did the outlet sell on that day?

- A 22000
- B 19000
- C 6000
- D 3000

Handwritten solution for Question 28:

Base exchange rates: A → 100k, B → 120k, C → 1k. Buying rates: A → 95k, B → 114k, C → 0.95k. Selling rates: A → 110k, B → 132k, C → 1.1k.

	Bought			
A				
B				
C				

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

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Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 29

What percentage of kids from S were studying in P?

- A 37%
- B 6%
- C 79%
- D 56%



NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

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	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
Total	13500	2700	1800	18000

 VIDEO SOLUTION

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 30

If the ATM could serve only 10 customers with a stock of fifty 500 rupee notes and a sufficient number of notes of other denominations, what is the maximum number of customers among these 10 who could have given 500 rupee notes as their preferences?

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

500	200	100
50	1	1
10	2	3
9	1	5
	0	

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VIDEO SOLUTION

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Question 31

What is the number of 500 rupee notes required to serve 50 customers with 500 rupee notes as their preferences and another 50 customers with 100 rupee notes as their preferences, if the total number of notes to be dispensed is the smallest possible?

- A 900
- B 800
- C 750
- D 1400

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

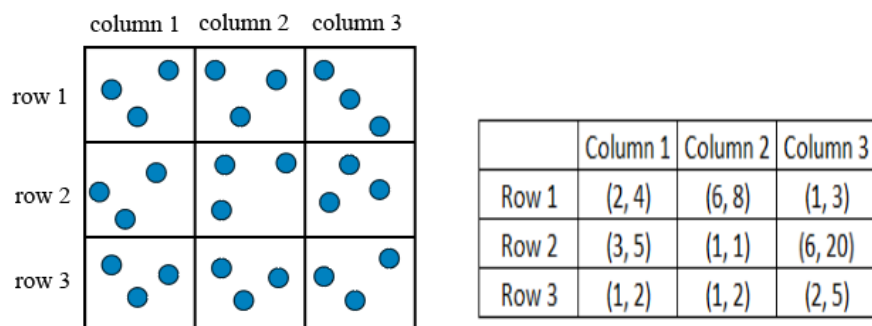
In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

500	200	100
50	1	1
10	2	3
9	1	5
	0	

cracku

VIDEO SOLUTION

Instructions



Three pouches (each represented by a filled circle) are kept in each of the nine slots in a 3×3 grid, as shown in the figure. Every pouch has a certain number of one-rupee coins. The minimum and maximum amounts of money (in rupees) among the three pouches in each of the nine slots are given in the table. For example, we know that among the three pouches kept in the second column of the first row, the minimum amount in a pouch is Rs. 6 and the maximum amount is Rs. 8.

There are nine pouches in any of the three columns, as well as in any of the three rows. It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. It is also known that the total amount of money kept in the three pouches in the first column of the third row is Rs. 4.

Question 32

The number of slots for which the total amount in its three pouches strictly exceeds Rs. 10 is

The video solution shows a hand-drawn diagram of the 3x3 grid and a table with the minimum and maximum amounts of money in rupees for each row and column. The table is identical to the one in the figure above.

VIDEO SOLUTION

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the

students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.

• If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 33

What is the minimum percentage of students who will vote in the election?

- A 32%
- B 40%
- C 38%
- D 36%

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A	R	TV	A	R
1	1	40%	20%	20%
1	2	60%	20%	4%

VIDEO SOLUTION

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Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 34

On a day of fluctuating market prices, the share price of XYZ Ltd. ends with a gain, i.e., it is higher at the close of the day compared to the opening value. Which trader got the maximum return on that day?

- A Bikram
- B Chetan
- C Abdul
- D Bikram or Chetan

E cannot be determined

[VIEW SOLUTION](#)

Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

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NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 35

In a follow up survey of the same kids two years later, it was found that all the kids were now in school. Of the kids who were not in school earlier, in one region, 25% were in G now, whereas the rest were enrolled in P; in the second region, all such kids were in G now; while in the third region, 50% of such kids had now joined G while the rest had joined P. As a result, in all three regions put together, 50% of the kids who were earlier out of school had joined G. It was also seen that no surveyed kid had changed schools.

What number of the surveyed kids now were in G in W?

- A 6000
- B 5250
- C 6750
- D 6300

cracku

NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. They found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers had dropped out of school before completing primary education or not. The following table gives the number of kids in different types of schools who dropped out of school before completing primary education:

	G	P	O	Total
NE	500	300	5000	
W	4200	1900	1200	7300
S	5100	300	300	5700
Total	13500	2700	1800	18000

VIDEO SOLUTION

Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- A and B are In the same row or column
- A is at a lower height than B
- If there is/are any individuals (s) between A and B, such individual(s) must be at a height lower than that of A.

Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 36

Which of the following is true for any individual at a platform of height 1 m in this layout?

- They can be reached by all the individuals in their own row and column.
- They can be reached by at least 4 individuals.
- They can be reached by at least one individual.
- They cannot be reached by anyone.

cracku

1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table above, consider the individual seated at height 8 in 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

VIDEO SOLUTION

Free Videos for CAT Preparation

Question 37

Which of the following statements is true about this layout?

- A Each row has an individual who can be reached by 5 or more individuals.
- B Each row has an individual who cannot be reached by anyone.
- C Each row has at least two individuals who can be reached by an equal number of individuals.
- D All individuals at the height of 9 m can be reached by at least 5 individuals.

cracku

1	6	1	2	4	3
2	9	5	3	2	8
3	7	8	4	6	5
4	3		5	1	2
5	1			3	9

Thus in the table above, consider the individual seated at height 8 in 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

VIDEO SOLUTION

Instructions

Directions for the following five questions: Answer the following questions based on the information given below:

Abdul, Bikram and Chetan are three professional traders who trade in shares of a company XYZ Ltd. Abdul follows the strategy of buying at the opening of the day at 10 am and selling the whole lot at the close of the day at 3 pm. Bikram follows the strategy of buying at hourly intervals: 10 am, 11am, 12 noon, 1 pm and 2 pm, and selling the whole lot at the close of the day. Further, he buys an equal number of shares in each purchase. Chetan follows a similar pattern as Bikram but his strategy is somewhat different. Chetan's total investment amount is divided equally among his purchases. The profit or loss made by each investor is the difference between the sale value at the close of the day less the investment in purchase. The "return" for each investor is defined as the ratio of the profit or loss to the investment amount expressed as a percentage.

Question 38

One day, two other traders, Dane and Emily joined Abdul, Bikram and Chetan for trading in the shares of XYZ Ltd. Dane followed a strategy of buying equal numbers of shares at 10 am, 11 am and 12 noon, and selling the same numbers at 1 pm, 2 pm and 3 pm. Emily, on the other hand, followed the strategy of buying shares using all her money at 10 am and selling all of them at 12 noon and again buying the shares for all the money at 1 pm and again selling all of them at the close of the day at 3 pm. At the close of the day the following was observed.

- Abdul lost money in the transactions.
- Both Dane and Emily made profits.
- There was an increase in share price during the closing hour compared to the price at 2 pm.
- Share price at 12 noon was lower than the opening price

Which of the following is necessarily false?

- A Share price was at its lowest at 2 pm
- B Share price was at its lowest at 11 am
- C Share price at 1 pm was higher than the share price at 2 pm
- D Share price at 1 pm was higher than the share price at 12 noon
- E more than one of the above

[VIEW SOLUTION](#)

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 39

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

[VIDEO SOLUTION](#)

CAT Percentile Predictor

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 40

The company is planning to launch a balanced diet required for growth needs of adolescent children. This diet must contain at least 30% each of carbohydrate and protein, no more than 25% fat and at least 5% minerals. Which one of the following combinations of equally mixed ingredients is feasible?

- A O and P
- B R and S
- C P and S
- D Q and R
- E O and S

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
B) Three
C) Two
D) None

Handwritten notes on the video solution:
Q & O: 10% protein, 30% protein
S & R: 10% protein, 30% protein
S & P: 10% protein, 30% protein

VIDEO SOLUTION

Instructions

A country has the following types of traffic signals.

3 red lights = stop

2 red lights = turn left

1 red light = turn right

3 green lights = go at 100 km/hr speed

2 green lights = go at 40 km/hr speed

1 green light = go at 20 km/hr speed

A motorist starts at a point on a road and follows all traffic signals. His car is heading towards the north. He encounters the following signals (the time mentioned in each case below is applicable after crossing the previous signal).

Starting point - 1 green light

After half an hour, 1st signal - 2 red and 2 green lights

After 15 min, 2nd signal - 1 red light

After half an hour, 3rd signal - 1 red and 3 green lights

After 24 min, 4th signal - 2 red and 2 green lights

After 15 min, 5th signal - 3 red lights

Question 41

The total distance travelled by the motorist from the starting point till the last signal is

- A 90 km
- B 100 km
- C 120 km
- D None of these

[VIEW SOLUTION](#)

Instructions

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers dropped out of school before completing primary education or not.. The table below gives the number of kids in different types of schools for mothers who dropped out of school before completing primary education:

	G	P	O	Total
NE	4200	500	300	5000
W	4200	1900	1200	7300
S	5100	300	300	5700
TOTAL	13500	2700	1800	18000

It is also known that:

1. In S, 60% of the surveyed kids were in G. Moreover, In S, all surveyed kids whose mothers had completed primary education were in school.
2. In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
3. The number of kids in G in NE was the same as the number of kids in G in W.

Question 42

In a follow up survey of the same kids two years later, it was found that all the kids were now in school. Of the kids who were not in school earlier, in one region, 25% were in G now, whereas the rest were enrolled in P; in the second region, all such kids were in G now; while in the third region, 50% of such kids had now joined G while the rest had joined P. As a result, in all three regions put together, 50% of the kids who were earlier out of school

had joined G. It was also seen that no surveyed kid had changed schools.
What percentage of the surveyed kids in S, whose mothers had dropped out before completing primary education, were in G now?

- A 94.7%
- B 89.5%
- C 93.4%
- D Cannot be determined from the given information

cracku

NE = 7500 W = 12500 S = 10,000

A study to look at the early learning of rural kids was carried out in a number of villages spanning three states, chosen from the North East (NE), the West (W) and the South (S). 50 four-year old kids each were sampled from each of the 150 villages from NE, 250 villages from W and 200 villages from S. It was found that of the 30000 surveyed kids 55% studied in primary schools run by government (G), 37% in private schools (P) while the remaining 8% did not go to school (O).

The kids surveyed were further divided into two groups based on whether their mothers had dropped out of school before completing primary education or not. The following table gives the number of kids in different types of schools who dropped out of school before completing primary education:

	G	P	O	Total
NE	500	300	5000	
W	4200	1900	1200	7300
S	5100	300	300	5700
Total	13500	2700	1800	18000

VIDEO SOLUTION

Important Verbal Ability Questions for CAT (Download PDF)

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 43

The value of the assets distributed among Neeta, Seeta and Geeta was in the ratio of 1:2:3, while the gold coins were distributed among them in the ratio of 2:3:4. One child got all three flats and she did not get the house.

One child, other than Geeta, got Rs. 30 lakh in bank deposits.

How much did Geeta get in bank deposits (in lakhs of rupees)?

cracku

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children, Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child, a flat went to one child, and similarly, a gold coin went to one child.

Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

Handwritten notes on the screen:

Assets: 70 - BD, 50 - House, 3x30 - Flats, X - Gold Coins

210

Handwritten calculations on the screen:

N 70, S 70, G 70

VIDEO SOLUTION

Instructions

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

A currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The following facts are known about the outlet on a particular day:

1. The amount of L used by the outlet to buy C equals the amount of L it received by selling C.
2. The amounts of L used by the outlet to buy A and B are in the ratio 5:3.
3. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.
4. The outlet received 88000 units of L by selling A during the day.
5. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.
6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 51000 units of C.

Question 44

What was the buying exchange rate of currency C with respect to currency L on that day?

- A 1.10
- B 0.95
- C 2.20
- D 1.90

cracku

The base exchange rate of a currency X with respect to a currency Y is the number of units of currency Y which is equivalent in value to one unit of currency X. Currency exchange outlets buy currency at buying exchange rates that are lower than base exchange rates, and sell currency at selling exchange rates that are higher than base exchange rates.

An international currency exchange outlet uses the local currency L to buy and sell three international currencies A, B, and C, but does not exchange one international currency directly with another. The base exchange rates of A, B and C with respect to L are in the ratio 100:200:1. The buying exchange rates of each of A, B, and C with respect to L are 5% below their corresponding base exchange rates, and their selling exchange rates are 10% above their corresponding base exchange rates. The exchange rates are known at the outlet on a particular day:

1. The amount of L received by the outlet to buy C equals the amount of L it received by the outlet to buy A and B are in the ratio 5:3.

2. The amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

3. The outlet received 8800 units of L by selling A during the day.

4. The outlet started the day with some amount of L, 2500 units of A, 4800 units of B, and 48000 units of C.

5. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 48000 units of C.

6. The outlet ended the day with some amount of L, 3300 units of A, 4800 units of B, and 48000 units of C.

Handwritten notes on the left:

A → 100k. 95k. 110k.
B → 200k. 114k. 132k.
C → 1k. 0.95k. 1.1k.
Base Buying Selling

	Bought				
A					
B					
C					

VIDEO SOLUTION

Instructions

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

Question 45

What is the maximum number of customers that the ATM can serve with a stock of fifty 500 rupee notes and a sufficient number of notes of other denominations, if all the customers are to be served with at most 20 notes per withdrawal?

- A 12
- B 10
- C 13
- D 16

cracku

An ATM dispenses exactly Rs. 5000 per withdrawal using 100, 200 and 500 rupee notes. The ATM requires every customer to give her preference for one of the three denominations of notes. It then dispenses notes such that the number of notes of the customer's preferred denomination exceeds the total number of notes of other denominations dispensed to her.

In how many different ways can the ATM serve a customer who gives 500 rupee notes as her preference?

Handwritten calculations on the right:

500 500 200 100
10 10 2 1
9 1 3
0 5

VIDEO SOLUTION

Data Interpretation for CAT Questions (download pdf)

Instructions

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous

campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

Question 46

What is the maximum possible voting margin with which one of the candidates can win?

- A 20%
- B 29%
- C 28%
- D 26%

Two students, Amiya and Ramya are the only candidates in an election for the position of class representative. Students will vote based on the intensity level of Amiya's and Ramya's campaigns and the type of campaigns they run. Each campaign is said to have a level of 1 if it is a staid campaign and a level of 2 if it is a vigorous campaign. Campaigns can be of two types, they can either focus on issues, or on attacking the other candidate.

If Amiya and Ramya both run campaigns focusing on issues, then

- The percentage of students voting in the election will be 20 times the sum of the levels of campaigning of the two students. For example, if Amiya and Ramya both run vigorous campaigns, then $20 \times (2+2)\%$, that is, 80% of the students will vote in the election.
- Among voting students, the percentage of votes for each candidate will be proportional to the levels of their campaigns. For example, if Amiya runs a staid (i.e., level 1) campaign while Ramya runs a vigorous (i.e., level 2) campaign, then Amiya will receive $1/3$ of the votes cast, and Ramya will receive the other $2/3$. The above-mentioned percentages change as follows if at least one of them runs a campaign attacking their opponent.
- If Amiya runs a campaign attacking Ramya and Ramya runs a campaign focusing on issues, then 10% of the students who would have otherwise voted for Amiya will vote for Ramya, and another 10% who would have otherwise voted for Amiya, will not vote at all.
- If Ramya runs a campaign attacking Amiya and Amiya runs a campaign focusing on issues, then 20% of the students who would have otherwise voted for Ramya will vote for Amiya, and another 5% who would have otherwise voted for Ramya, will not vote at all.
- If both run campaigns attacking each other, then 10% of the students who would have otherwise voted for them had they run campaigns focusing on issues, will not vote at all.

	A	R	IV
1	1	2	40% 20% 20%
2	2	1	60% 20% 40%

VIDEO SOLUTION

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 47

In what proportion P, Q and S should be mixed to make a diet having at least 60% carbohydrate at the lowest per unit cost?

- A 2 : 1 : 3
- B 4 : 1 : 2
- C 2 : 1 : 4
- D 3 : 1 : 2
- E 4 : 1 : 1

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

For a recuperating patient, the doctor recommended a diet containing at least 30% protein. In how many different ways can we prepare this diet by mixing at least two ingredients?

A) One
C) Three
E) None

Handwritten notes: Q & O mixed 10% 30% Protein. Q, S, R, P < 10%.

VIDEO SOLUTION

Instructions

Question Numbers: (55 to 58)

In a square layout of site 5m ~ 5m 25 equal-sized square platforms of different heights are built. The heights (in metre) of individual platforms are as shown below:

6	1	2	4	3
9	5	3	2	8
7	8	4	6	5
3	9	5	1	2
1	7	6	3	9

Individuals (all of same height) are seated on these platforms. We say an individual A can reach individual B, if all the three following conditions are met;

- (i) A and B are in the same row or column
- (ii) A is at a lower height than B
- (iii) If there is/are any individual(s) between A and B, such individual(s) must be at a height lower than that of A.

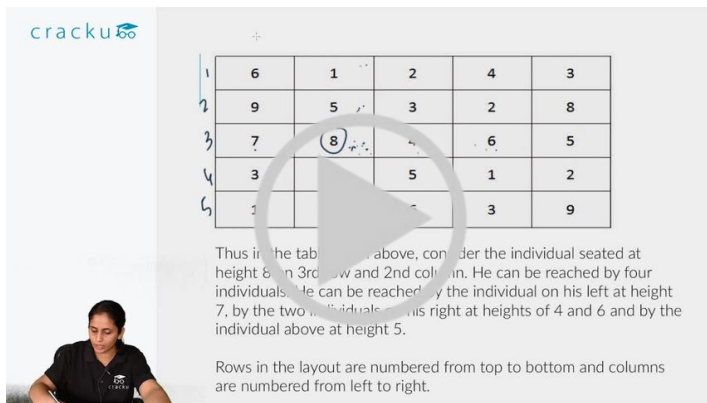
Thus in the table given above, consider the Individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

Question 48

How many individuals in this layout can be reached by just one individual?

- A 3
- B 5
- C 7
- D 8



Thus in the table given above, consider the individual seated at height 8 on 3rd row and 2nd column. He can be reached by four individuals. He can be reached by the individual on his left at height 7, by the two individuals on his right at heights of 4 and 6 and by the individual above at height 5.

Rows in the layout are numbered from top to bottom and columns are numbered from left to right.

VIDEO SOLUTION

Logical Reasoning for CAT Questions (download pdf)

Instructions

Out of 10 countries – Country 1 through Country 10 – Country 9 has the highest gross domestic product (GDP), and Country 10 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the GDP per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question 49

Which one among the countries 1, 4, 5, and 7 will have the largest population in 2027?

- A Country 1
- B Country 5
- C Country 7
- D Country 4

Out of 10 countries – Country 3 through Country 10 – Country 9 has the highest GDP per capita. GDP per capita is the GDP of a country divided by its population. The table below provides the following data about Country 1 through Country 8 for the year 2024.

- Column 1 gives the country's identity.
- Column 2 gives the country's GDP as a fraction of the GDP of Country 9.
- Column 3 gives the country's GDP per capita as a fraction of the per capita of Country 10.
- Column 4 gives the country's annual GDP growth rate.
- Column 5 gives the country's annual population growth rate.

Country	GDP	GDP per capita	GDP growth rate	Pop. growth rate
Country 1	0.15	0.25	2.5%	1.5%
Country 2	0.20	0.30	3.0%	2.0%
Country 3	0.25	0.35	3.5%	2.5%
Country 4	0.30	0.40	4.0%	3.0%
Country 5	0.35	0.45	4.5%	3.5%
Country 6	0.40	0.50	5.0%	4.0%
Country 7	0.45	0.55	5.5%	4.5%
Country 8	0.50	0.60	6.0%	5.0%

Assume that the GDP growth rates and population growth rates of the countries will remain constant for the next three years.

Question (1)

Which one among the countries 1 through 8, has the smallest population in 2024?

A) Country 8
B) Country 3
C) Country 7
D) Country 5

Population ~ GDP

VIDEO SOLUTION

Instructions

Directions for the following four questions:

A health-drink company's R&D department is trying to make various diet formulations, which can be used for certain specific purposes. It is considering a choice of 5 alternative ingredients (O, P, Q, R, and S), which can be used in different proportions in the formulations.

The table below gives the composition of these ingredients. The cost per unit of each of these ingredients is O: 150, P: 50, Q: 200, R: 500, S: 100.

Composition				
Ingredient	Carbohydrate %	Protein %	Fat %	Minerals %
S	45	50	0	5
R	5	50	40	5
Q	10	30	50	10
P	80	20	0	0
O	50	30	10	10

Question 50

Which among the following is the formulation having the lowest cost per unit for a diet having 10% fat and at least 30% protein? The diet has to be formed by mixing two ingredients.

- A P and Q
- B P and S
- C P and R
- D Q and S
- E R and S

There are only four brands of entry level smartphones called Azra, Bysi, Cxqi, and Dipq in a country. Details about their market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Azra	40	15,000	10
Bysi	25	20,000	30
Cxqi	15	30,000	40
Dipq	20	25,000	30

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqi offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqi came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

Question 52

The brand that had the highest profit in 2016 is:

- A Bysi
- B Dipq
- C Cxqi
- D Azra

VIDEO SOLUTION

Question 53

The complete list of brands whose profits went up in 2017 from 2016 is:

- A Azra, Bysi, Dipq
- B Cxqi, Azra, Dipq
- C Azra, Bysi, Cxqi
- D Bysi, Cxqi, Dipq

Details about the market share, unit selling price, and profitability (defined as the profit as a percentage of the revenue) for the year 2016 are given in the table below:

Brand	Market Share (%)	Unit Selling Price (Rs.)	Profitability (%)
Brand A	25	15,000	30
Brand B	35	10,000	20
Brand C	40	8,000	15

In 2017, sales volume of entry level smartphones grew by 40% as compared to that in 2016. Cxqj offered a 40% discount on its unit selling price in 2017, which resulted in a 15% increase in its market share. Each of the other three brands lost 5% market share. However, the profitability of Cxqj came down to half of its value in 2016. The unit selling prices of the other three brands and their profitability values remained the same in 2017 as they were in 2016.

2017 → 140

A.
B.
C.
D.

VIDEO SOLUTION

Instructions

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
- (b) 1 house worth Rs. 50 lakh
- (c) 3 flats, each worth Rs. 30 lakh
- (d) Certain number of gold coins, each worth Rs. 1 lakh

She wanted to distribute her assets among her three children; Neeta, Seeta and Geeta.

The house, any of the flats or any of the coins were not to be split. That is, the house went entirely to one child; a flat went to one child and similarly, a gold coin went to one child.

Question 54

The value of the assets distributed among Neeta, Seeta and Geeta was in the ratio of 1:2:3, while the gold coins were distributed among them in the ratio of 2:3:4. One child got all three flats and she did not get the house.

One child, other than Geeta, got Rs. 30 lakh in bank deposits.

How many gold coins did the old woman have?

- A 72
- B 90
- C 180
- D 216

cracku

An old woman had the following assets:

- (a) Rs. 70 lakh in bank deposits
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Question (1)

Among the three, Neeta received the least amount in bank deposits, while Geeta received the highest. The value of the assets was distributed equally among the children, as were the gold coins.

How much did Seeta receive in bank deposits (in lakhs of rupees)?

N 10
S 70
G 10

70 - BD
50 - House
3 × 30 - Flats
X - Gold Coins
210

VIDEO SOLUTION

Know the CAT Percentile Required for IIM Calls